

THÈSE

Pour obtenir le grade de

DOCTEUR DE L'UNIVERSITÉ GRENOBLE ALPES

École doctorale : EEATS - Electronique, Electrotechnique, Automatique, Traitement du Signal

Spécialité : Nano électronique et Nano technologies

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Conception d'un Système Mémoire pour Traitement de Données Creuses

Design of a Memory System for Sparse Data Processing

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THESIS

DESIGN OF A MEMORY SUB-SYSTEM FOR SPARSE DATA PROCESSING

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Résumé

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Abstract

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List of Publications

This thesis is based on the following publications:

- I Valentin Isaac--Chassande, Adrian Evans, Yves Durand, and Frédéric Rousseau. 2024. Dedicated Hardware Accelerators for Processing of Sparse Matrices and Vectors: A Survey. *ACM Trans. Archit. Code Optim.* 21, 2, Article 27 (Feb. 2024), 26 pages. Reference [5]
- II Valentin Isaac--Chassande, Adrian Evans, Yves Durand, and Frédéric Rousseau. 2024. SpDCache: Region-Based Reduction Cache for Outer-Product Sparse Matrix Kernels. In *2024 IEEE 35th International Conference on Application-specific Systems, Architectures and Processors (ASAP)*. IEEE Computer Society, Los Alamitos, CA, USA, 3–7. Reference [6]
- III **TODO**

The papers will be referred to in the thesis using their Roman numerals.

Summary

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Notations

$$\forall (a, b) \in \mathbb{Z}^2, \llbracket a, b \rrbracket = [a, b] \cap \mathbb{Z}$$

TODO

Chapter 1

Plan (Brouillon)

- Introduction (~3 p.)
 - "light" > give the entry point on sparse matrices and irregular accesses (1-2 p.)
 - introduce the structure of the manuscript (1 p.)
- Background (~5 p.)
 - irregular accesses -> def, issues (in memory but not only?), domains/applications (including sparse matrix processing) (1 p.)
 - Focus on sparse matrices, context: it is widely used, in many domains, sparse, large, structures
 - Sparse matrices in a kernel: show where the accesses are regular vs irregular (infos on basic sparse formats, basic kernels, basic algos...) (2-3 p.)
 - Hardware challenges clearly defined (gather, scatter...) (1 p.)
- SoA (~10 p.)
 - Present solutions in SW for CPU/GPU, such as special sparse formats, possibility to play on algo (gust analysis), tiling methods adapted for sparse data (specific sparse format + specific algos) (1 p.)
 - Present solutions in HW (dedicated accelerators) -> classification along common characteristics (1-2 p.)
 - Presentation of the major accelerators (selection of 3-4 important) (3-4 p.)
 - Problem of comparison / comparison of standalone / other analysis (issue of matrix structure) (1 p.)
 - Takeaways -> paths for designing an accelerator (end of SoA paper) (1-2 p.)
 - We choose a path -> show which one (coarse grain, at the level of the SoA, comparable to SortCache, PHI...)
- Conceptual solution (~10 p.)
 - Remainder of the path we choose with more details: cache / reductions / OP SpMV / ... (1 p.)
 - Emphasis the comparison between concerned SoA accelerators -> show what is already done and useful, show what is missing
 - Present the case of banded matrices as an example of structure to leverage (1-2 p.)
 - SpDCache, architecture presentation with two regions, details on REDs transactions, IBRC vs OBRT (7 p.)
- Theory (~23 p.)
 - Explain the objectives of this chapter (proof of concept + dimensioning) (1

- p.)
- Describe the design space of the path we choose (with remainders if necessary) -> introduce base parameters and/or variables, give the exact algorithm... (1 p.)
- Show the interest of separate cache regions for A/b and c (formal) -> show that a GC is obviously not efficient when random accesses / confirm why SortCache, PHI, ... exist (2 p.)
- Show that A and b are not a limiting factor with a formal reasoning -> give the needed space for GRC when prefetcher (2-3 p.)
- We use a GRC (no modification) for A and b -> get nb loads and writes, hit, miss, evict (should be all hits, and one traversal only)
- Show the interest of a reduction cache region for c -> REDs / reduce the number of mem accesses / link to SoA (see ASAP) (2 p.)
- We created a model/simulation in C of the reduction cache / validation of the model (1-2 p.)
- Show that reduction cache does not handle randomness -> model graph (1-2 p.)
- Show RedC 1reg vs RedC 2reg on banded example (2-3 p.)
- Show best cache dimensions for the dense part (band) (1 p.)
- Show that sparse region (out of band) could use an other region for fine grain density (case of sub bands) -> dimensions (1-2 p.)
- Show that sparse region could use adapted HW because of the isolated elements -> dimensions (1-2 p.)
- Show how it would work with a "full random" matrix -> how does it affects the dimensions (1 p.)
- Conclude on our solution and dimensioning (1 p.)
- Hardware solution (~10 p.)
 - Present our hardware solution, based on the concept
 - Start with basic info on HPDcache (only the one that are useful for SpDCache)
 - Describe modif in pipeline / modif in RAM accesses ...
- Experimental results (~10 p.)
 - Experimentations -> matrices chosen, generated / metrics / ...
 - Results / comparison with theory / analysis / conclusion
- Alternate solutions (~8 p.)
 - Architecture presentation with three regions (add OBRC, CB...) (2 p.)
 - Architecture of eviction anticipation (2 p.)
 - Present multicore version -> scale (2 p.)
 - Present additional optimisation (region config, learn from first iteration) (1-2 p.)
- Annexes
 - All sparse formats
 - Other accelerators of SoA
 - Model of the reduction cache

Chapter 2

SoA

...

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Conclusion

1 TODO

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Appendices

A TODO

TODO if necessary

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List of Figures

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List of Algorithms

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List of Source Code

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